Bangladesh University of Business & Technology (BUBT)

Lab Report

Course Title : Md. Takwat Hossain

Course Code : CSE102

Lab Report No : 01

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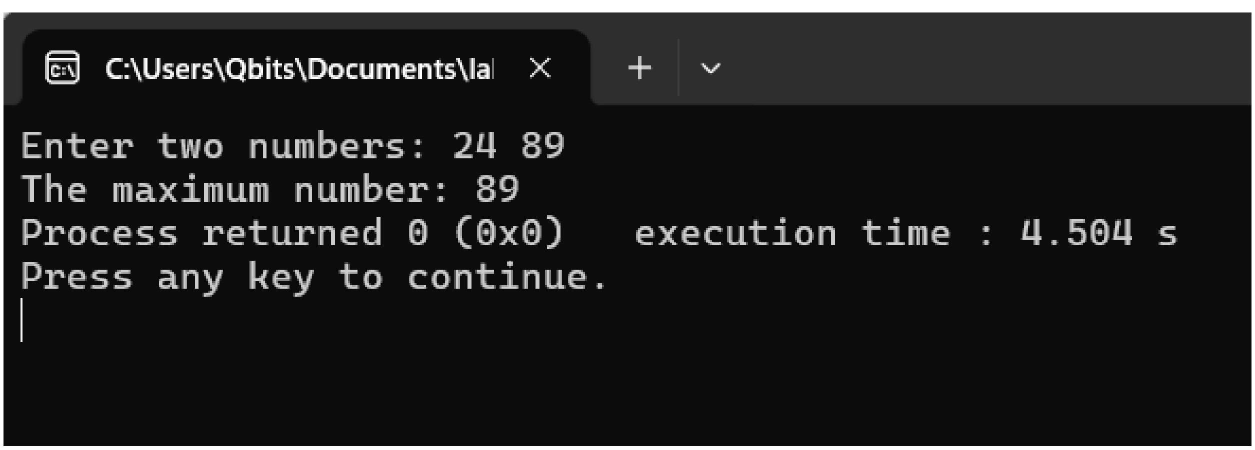
Question 01 :Write a C program to find maximum between two numbers.

```
#include<stdio.h>

int main(){

int x,y;

printf("Enter two numbers: ");

scanf("%d %d",&x,&y);

if(x>y){

   printf("The maximum number: %d",x);

}

else{

   printf("The maximum number: %d",y);

}

return 0;

}
```

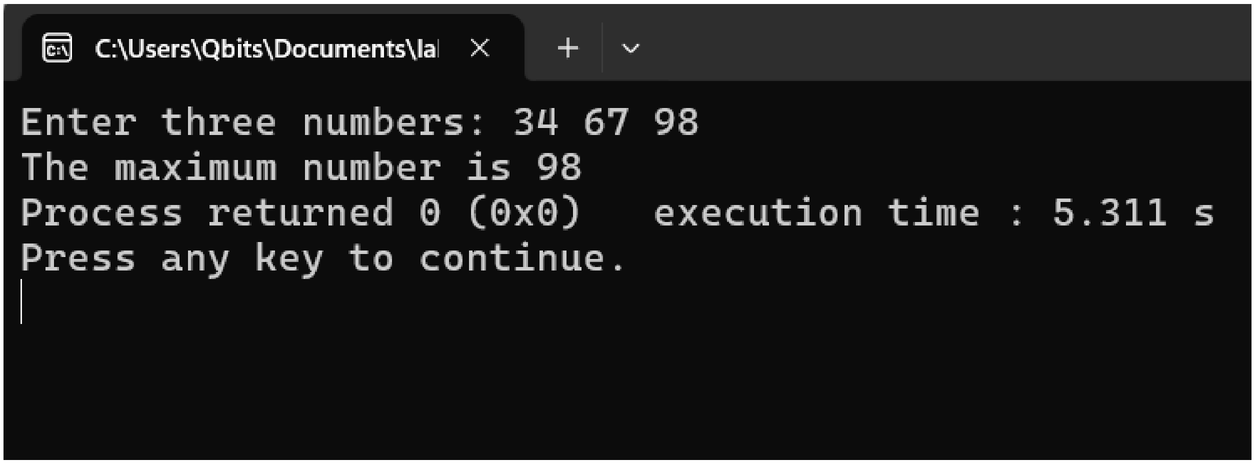
Output:



Question 2: Write a C program to find maximum between three numbers.

```
#include<stdio.h>

int main(){

int x,y,z;

printf("Enter three numbers: ");

scanf("%d%d%d",&x,&y,&z);

if(x>=y && x>=z){

   printf("The maximum number %d",x);

}

else if(y>=x && y>=z){

   printf("The maximum number %d",y);

}

else{

   printf("The maximum number is %d",z);

}


return 0;

}
```

Output:

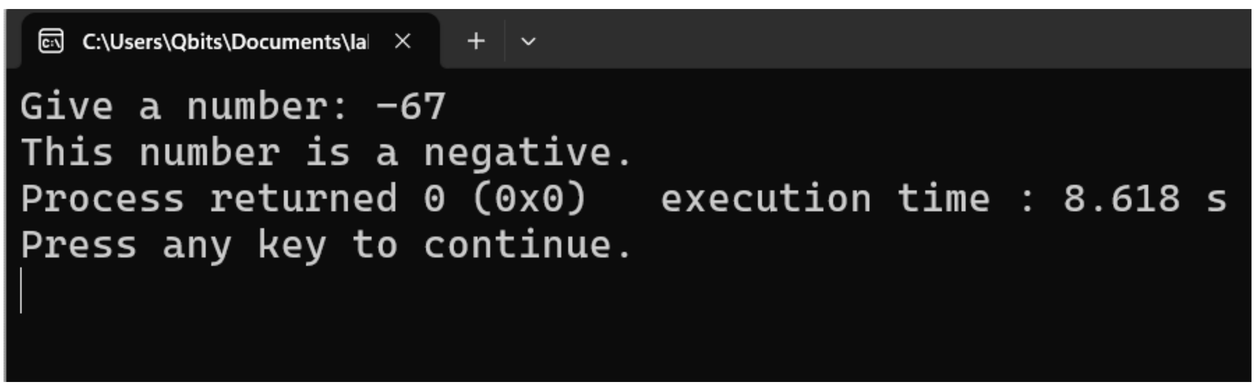


Question 3: Write a C program to check whether a number is negative, positive or zero.

```
#include<stdio.h>

int main(){

int x;

printf("Give a number: ");

scanf("%d",&x);

if(x>0){

   printf("This number is positive.");

}
```

```
else if(x<0){
    printf("This number is a negative.");
}
else{
    printf("This number is zero.");
}
return 0;
}
```

Output:



Question 4: Write a C program to check whether a number is divisible by 5 and 11 or not.

```
#include<stdio.h>
```

```c
int main(){

int x;

printf("Give a number: ");

scanf("%d",&x);

if(x%5==0 && x%11==0){

   printf("This number is divided by 5 and 11.");

}

else{

   printf("This number is not devided by 5 and 11.");

}

return 0;

}
```

Output :

```
C:\Users\Qbits\Documents\la

Give a number: 110
This number is divided by 5 and 11.
Process returned 0 (0x0)   execution time : 4.035 s
Press any key to continue.
```

Question 5: Write a C program to check whether a number is even or odd .

```
#include<stdio.h>
int main(){
int x;
printf("Give a number: ");
scanf("%d",&x);
if(x%2==0){
   printf("This number is an even number.");
}
else{
   printf("This number is an odd number.");
}


return 0;
}
```

Output :

```
Give a number: 48
This number is an even number.
Process returned 0 (0x0)   execution time : 2.273 s
Press any key to continue.
```

Question 6: Write a C program to check whether a year is leap year or not.

```
#include<stdio.h>

int main(){

int x;

printf("Enter a year: ");

scanf("%d",&x);

if((x%400==0) || (x%4==0 && x%100!=0)){

   printf("This year is a leap year.");

}

else{

   printf("This year is not a leap year.");

}

return 0;

}
```

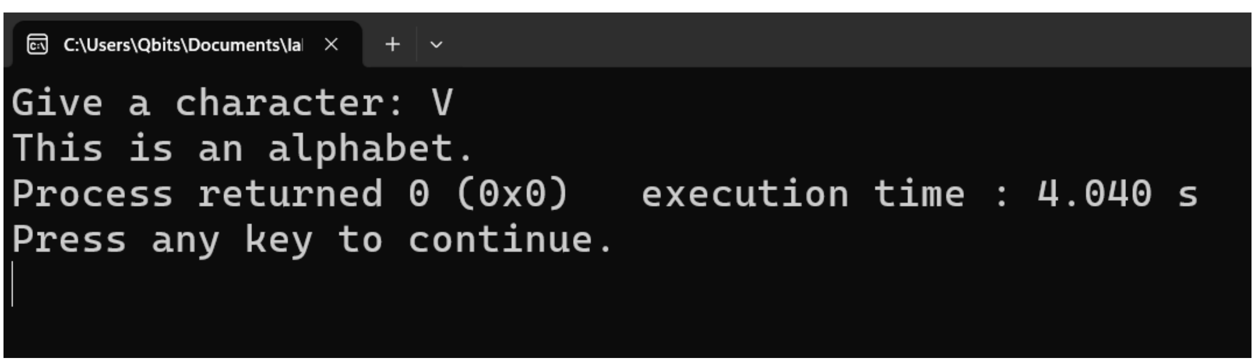
Output:

```
C:\Users\Qbits\Documents\la

Enter a year: 2038
This year is not a leap year.
Process returned 0 (0x0)   execution time : 6.404 s
Press any key to continue.
```

Question 7: Write a C program to check whether a character is alphabet or not.

```
#include<stdio.h>

int main(){

char x;

printf("Give a character: ");

scanf("%c",&x);

if((x>='a' && x<='z') || (x>='A' && x<='Z')){

   printf("This is an alphabet.");

}

else{

   printf("This is not an alphabet.");

}


return 0;

}
```

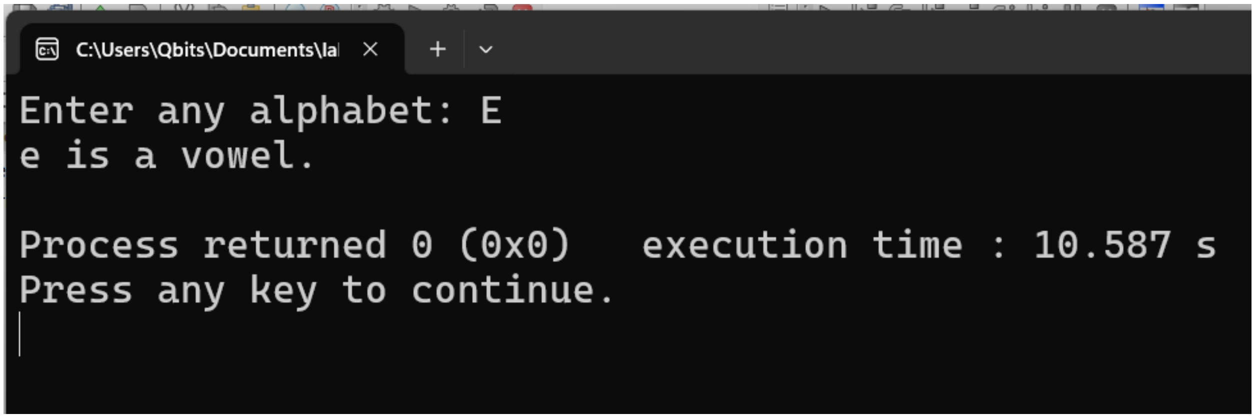
Output:



Question 8: Write a C program to input any alphabet and check whether it is vowel or consonant.

```
#include <stdio.h>


int main() {
    char ch;
  printf("Enter any alphabet: ");
    scanf("%c", &ch);
ch = tolower(ch);
    if (ch >= 'a' && ch <= 'z') {
        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
            printf("%c is a vowel.\n", ch);
        } else {
            printf("%c is a consonant.\n", ch);
        }
    } else {
        printf("Invalid input! Please enter an alphabet.\n");
    }
    return 0;
}
```

Output:

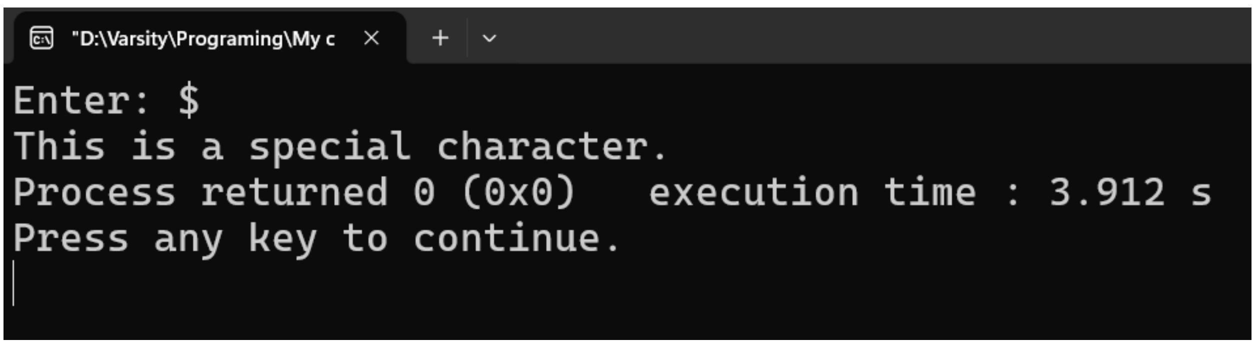


Question 9: Write a C program to input any character and check whether it is alphabet, digit or special character.

```
#include<stdio.h>

int main(){

char x;

printf("Enter: ");

scanf("%c",&x);

if(x>='A' && x<='Z'){

   printf("This is an uppercase letter.");

}

else if(x>='a' && x<='z'){

   printf("This is a lowercase letter.");

}
```

```
else if(x>='0' && x<='9'){
    printf("This is a digit");
}
else{
    printf("This is a special character.");
}
return 0;
}
```

Output:



Question 10: Write a C program to check whether a character is uppercase or lowercase alphabet.

```
#include <stdio.h>

int main() {
    char ch;
```

```c
    printf("Enter a character: ");
    if (scanf(" %c", &ch) != 1) {
        printf("Error reading input.\n");
        return 1;
    }

    if (ch >= 'A' && ch <= 'Z') {
        printf("\nResult:\n");
        printf("The character '%c' is an UPPERCASE alphabet.\n", ch);
    }
    else if (ch >= 'a' && ch <= 'z') {
        printf("\nResult:\n");
        printf("The character '%c' is a LOWERCASE alphabet.\n", ch);
    }
    else {
        printf("\nResult:\n");
        printf("The character '%c' is not a standard English alphabet (A-Z or a-z).\n", ch);
    }

    return 0;
}
```

Output:

"D:\Varsity\Programing\My c

```
Enter a character: Q

Result:
The character 'Q' is an UPPERCASE alphabet.

Process returned 0 (0x0)   execution time : 2.821 s
Press any key to continue.
```